Mid Semester Examination – 2020

School of Computer Engineering

Kalinga Institute of Industrial Technology (KIIT) Deemed to be University

Subject: Introduction to Operating System , Code :CS - 3044

Time: 1.5 Hrs | Full Marks: 20

(Answer any Four Questions including Question No. 1)

1. Answer the following questions briefly. [1 X 5]
   a. What is operating system and what are the goals of operating system ?
   b. What is the difference between preemptive and non preemptive scheduling ?
   c. What is the difference between process and program ?
   d. What is the work of medium term scheduler ?
   e. What is process synchronization ?
2. Compare among Batch, Multi programming, Time sharing and Multi tasking operating system. [5]
3. Consider the following snapshot of the system: [5]

Process	Burst Time (ms)	Arrival Time
P1	10	0
P2	5	1
P3	2	2
P4	1	2
P5	7	3

Using shortest remaining time first scheduling and first come first serve, find the waiting time and turn around time of each process. Also draw the Gantt chart for each scheduling algorithm.

4. What are the various states of a process? Explain about the state transitions of a process during its life with state transition diagram. [5]
5. Explain Lock Variable mechanism and show why it is not an appropriate solution for process synchronization? [5]
6. Consider the following processes arrived in a system. [5]

Process	Next CPU Burst Time(ms)	Arrival Time
A	4	0
B	5	1
C	6	2
D	3	3
E	1	4
F	4	5

Calculate the average waiting time of the processes if the scheduling algorithm is Round Robin with time slice length as 2 ms.

~~~~~ ALL THE BEST ~~~~~
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